

MEMORANDUM

To: CEO & CFO, Tenacious Intelligence Corporation | From: Yakob Dereje, Research Partner | May 2, 2026
Re: Tenacious-Bench v0.1 — Sales Agent Evaluation & LoRA Adapter Results

EXECUTIVE SUMMARY

We built **Tenacious-Bench v0.1** — a 293-task machine-verifiable benchmark targeting the gaps τ^2 -Bench retail cannot grade: tone compliance against the Tenacious style guide, signal grounding in public hiring data, and commercial safety in B2B cold outreach. A LoRA adapter (Qwen 2.5-0.5B, 2,541 augmented training pairs, \$0.00 on Colab T4) lifted pass@1 from 0.491 to 0.754 on the sealed 57-task held-out partition: **Delta A = +0.263, 95% CI [0.140, 0.386], p < 0.0001** (paired bootstrap, n = 10,000). The trained adapter also outperformed a carefully prompt-engineered baseline on the same backbone by **+0.140 (Delta B positive)** — the lift is attributable to learned weights, not prompting.

HEADLINE RESULTS

Metric	Result	Source
Baseline pass@1 (Week 10 agent)	0.491	ablation_results.json
Trained adapter pass@1	0.754	ablation_results.json
Delta A (trained vs. baseline)	+0.263	primary lift metric
95% CI for Delta A	[0.140, 0.386]	entirely positive — robust
p-value (paired bootstrap n=10,000)	< 0.0001	highly significant
Delta B (trained vs. prompt-engineered)	+0.140 $\square$	training beats PE
Delta C (vs. τ^2 -Bench 0.333) [info only]	+0.421	Week 10 score reused
Training cost	\$0.00	Colab T4, 19.1 min
Inference cost — baseline / trained	\$0.00012 / \$0.00012 per task	zero marginal cost delta

DELTA B — HONEST REPORT ON PROMPT-ENGINEERED BASELINE

The prompt-engineered baseline used the identical Qwen 2.5-0.5B backbone with the complete Tenacious style guide in the system prompt — the strongest possible prompt-only intervention. It scored 0.614. The trained adapter scored 0.754, a **+0.140 advantage for training**. Delta B is positive: the adapter has internalised style constraints in its weights, not merely followed instructions. At production scale this means shorter prompts, lower per-call token cost, and more consistent output under adversarial inputs.

COST-PARETO & PRODUCTION RECOMMENDATION

Cost delta: \$0.00 training (Colab T4, 19.1 min). Inference cost is \$0.00012/task with or without the adapter — zero marginal overhead. Latency is also unchanged: the adapter runs on the same Qwen 2.5-0.5B backbone, adding no additional inference steps; measured p50 latency remains consistent with the Week 10 baseline of 1,219ms per email composition call. A +26.3pp lift with zero cost change is an unambiguous Pareto improvement. **Recommendation: Deploy unconditionally.** Delta A is highly significant, Delta B is positive, the CI excludes zero, and cost is unchanged. The adapter replaces the email composition component in the next production deployment. Kill-switch conditions on Page 2 govern rollback.

SKEPTIC'S APPENDIX

FOUR BEHAVIORS WITH ZERO TASKS IN v0.1 (with v0.2 fix for each)

Multi-turn objection handling. Every v0.1 task is a single cold-outreach turn. Zero tasks test agent responses to objections, soft deferral, or re-engagement after a hard no. v0.2: Add 40–60 three-turn conversation trajectories graded at each reply step.

Reply classification accuracy. The Conversion Engine classifies replies into five categories and routes accordingly. v0.1 contains zero classification tasks. A hard_no misclassified as curious triggers SMS to a hostile prospect. v0.2: Add 50–75 classification tasks per category with automated label checking.

Zero-signal fabrication. Every v0.1 task supplies at least one hiring signal. No task tests whether the agent invents signals when given an empty brief — the highest-risk production failure. v0.2: Add 25+ zero-signal adversarial tasks graded strictly on abstention vs. fabrication.

Segment 4 (AI capability gap) coverage. Segment 4 has 20 training pairs and zero dedicated held-out tasks — performance unvalidated. v0.2: Enforce minimum 15 held-out tasks per ICP segment with segment-stratified reporting.

GROUND TRUTH FAITHFULNESS — MEASUREMENT LOSSINESS ON EXISTING TASKS

The grounding check requires at least one signal from the hiring brief in the email body. Signals derive from **Crunchbase ODM** (1,001-company snapshot, Apache 2.0) and **layoffs.fyi** (2,361 records, April 2026) — public proxies, not live Tenacious intelligence. Three lossiness sources affect fidelity on tasks v0.1 already includes: **(1)** Crunchbase may lag real hiring velocity by 30–60 days, so grounded emails may cite stale signals; **(2)** the LLM tone judge is calibrated against only 12 hand-labeled examples — valid novel phrasing may be penalised; **(3)** the judge prompt contains no check for fabricated numeric claims, so TB-SG23-class violations pass silently. These sources bound score precision above 0.85.

ONE HONEST UNRESOLVED TRAINING FAILURE

Numeric fabrication alignment regression. On TB-SG23 — a hand-authored adversarial task containing an invented pricing claim (\$1.2M for a 12-month engagement) — the trained adapter scores 1.0 when the expected score is 0.0. The model satisfies every measurable style constraint but does not abstain from numeric invention. Root cause: the 2,541 training pairs consisted exclusively of correct emails as positive examples — no (fabricated-claim, rejection) pairs were included. This is a model behaviour failure, not a benchmark gap. Unresolved in v0.1. **Resolution:** add 30–50 DPO rejection-sampling pairs where fabricated pricing is the rejected output, then retrain.

KILL-SWITCH TRIGGER CONDITIONS

Trigger	Measurable Condition	Action
Style regression	pass@1 on Tenacious-Bench dev < 0.500 on any rolling 50-task window	Rollback adapter; revert to PE baseline
Banned phrase in production	Any outbound Resend email contains a phrase from the 40-item banned list	Immediate rollback + audit last 100 sends
Prospect quality signal	≥ 3 hard_no replies in any 7-day window cite "template", "generic", or "automated"	Suspend adapter; human review of 20 sends
Ablation regression	Delta A on refreshed held-out (≥ 15 tasks per ICP segment) is negative	No redeploy without DPO retraining